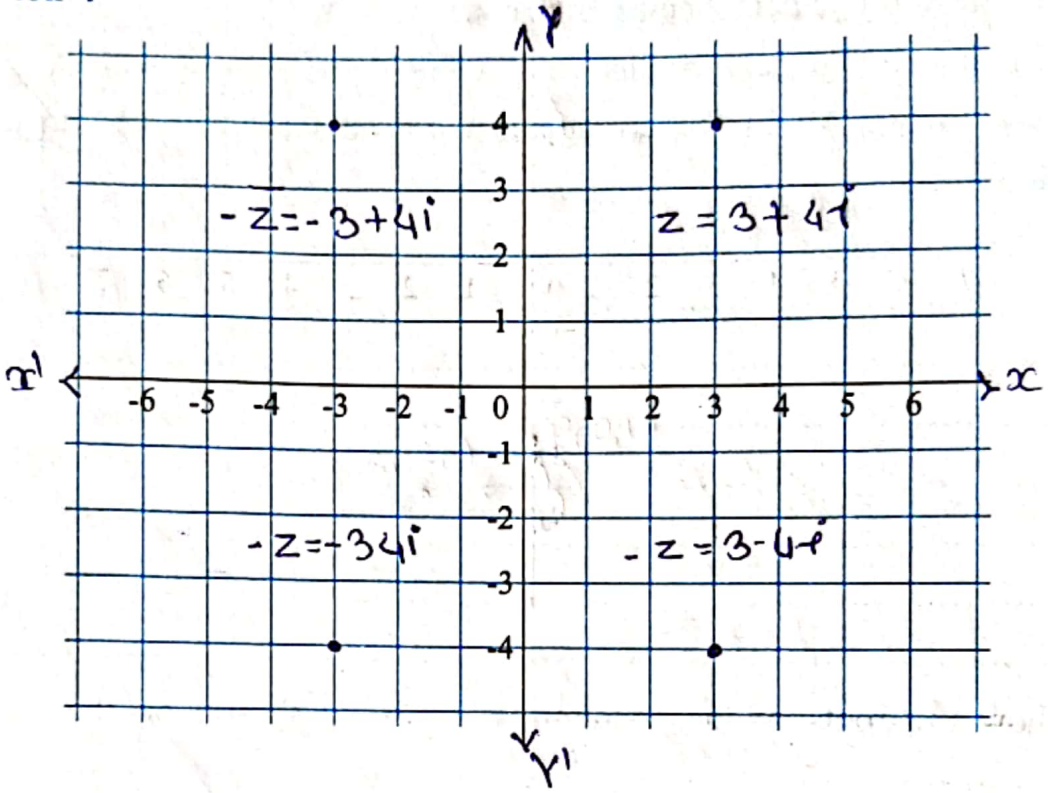
10. Complex Numbers

(1) If $z = 3 + 4i$ then show the following on the Argand diagram

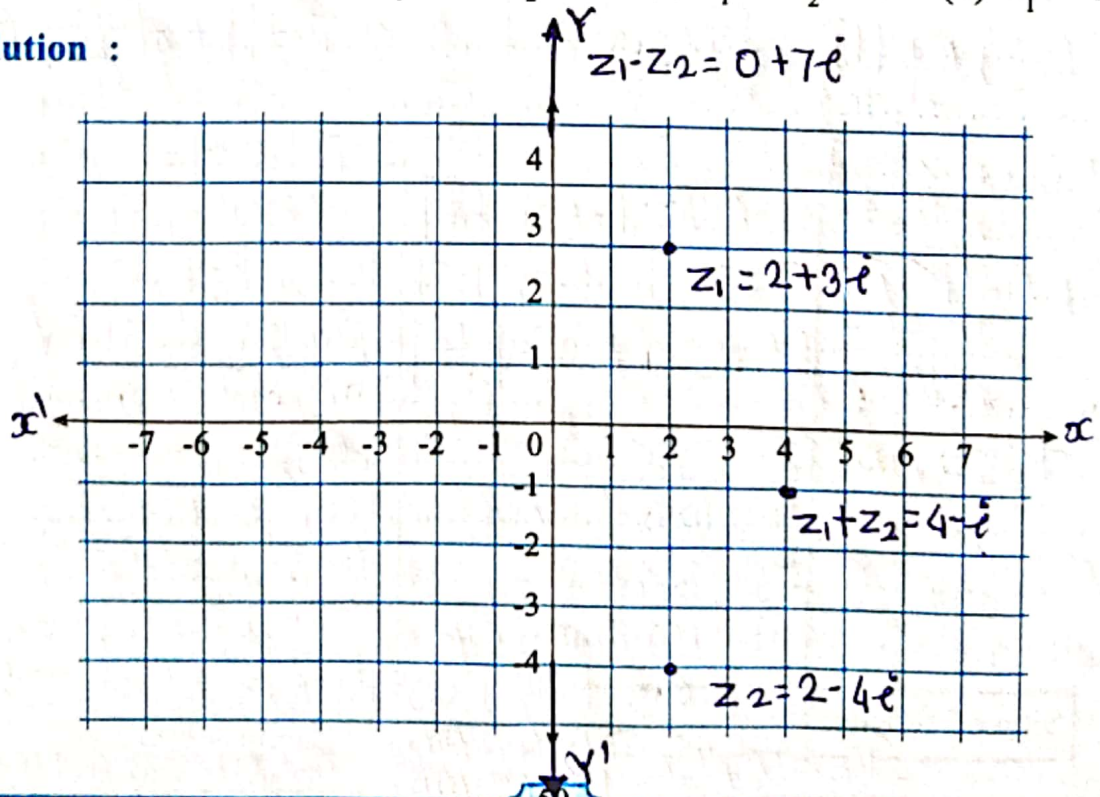
(a) z (b) $-z$ (c) $\bar{z}$ (d) $-\bar{z}$

Solution :



2) If $z_1 = 2 + 3i$ and $z_2 = 2 - 4i$ then show the following on the Argand diagram (a) z_1 (b) z_2 (c) $z_1 + z_2$ (d) $z_1 - z_2$

Solution :



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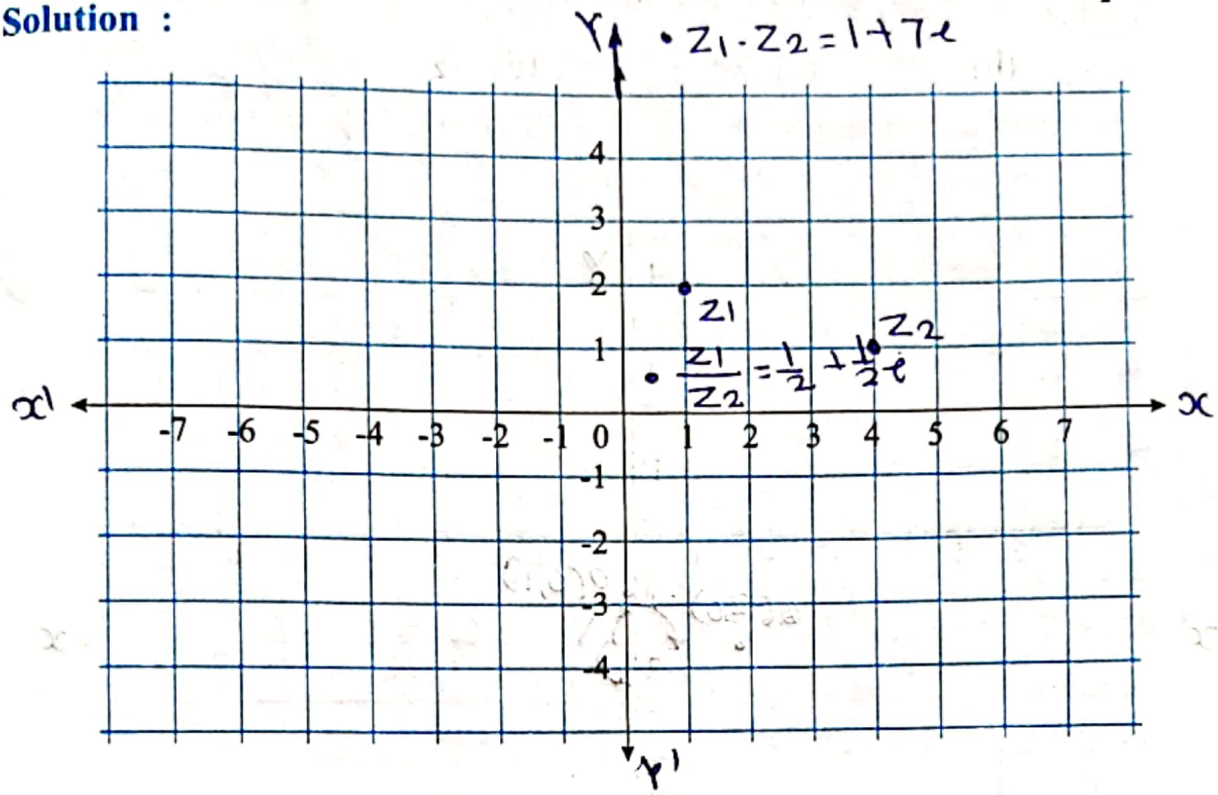
3) If $z_1 = 1 + 2i$ and $z_2 = 3 + i$. Show the following on the Argand diagram (a) z_1 (b) z_2 (c) $z_1 \cdot z_2$ (d) $\frac{z_1}{z_2}$

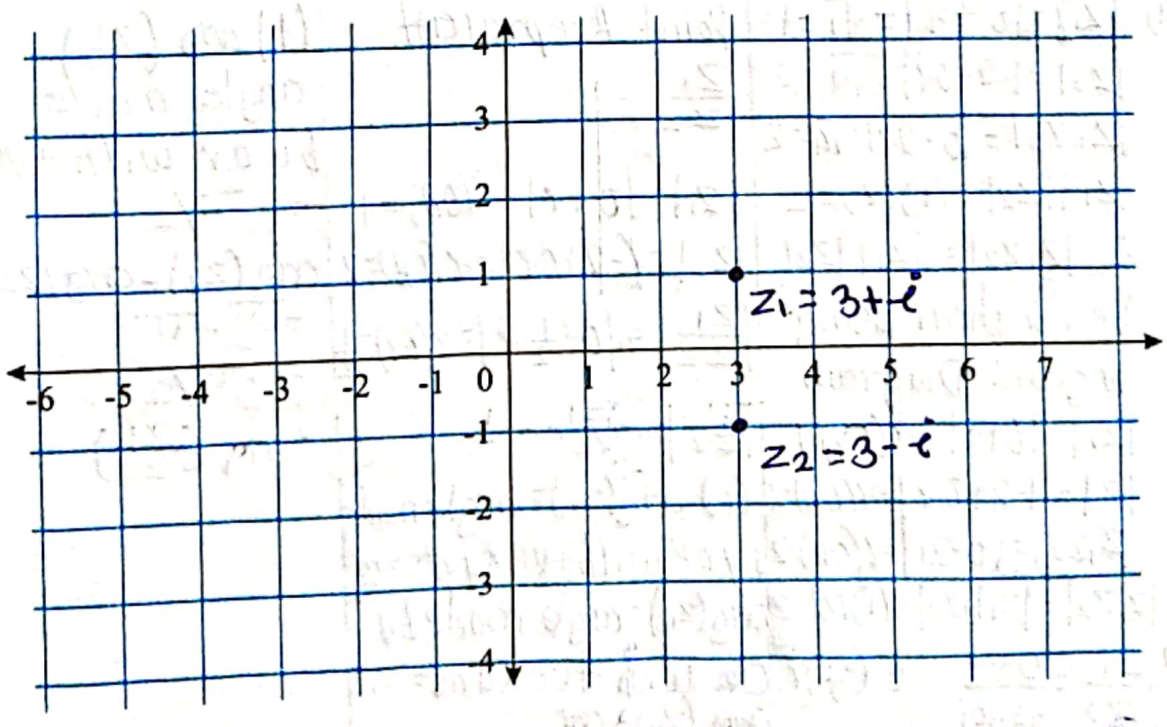
Solution :



4) Show the roots of the equation $x^2 - 6x + 10 = 0$ on the Argand diagram.

Solution : $x^2-6x+10=a, a=1, b=-6, c=10$ $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = 3 \pm i$

roots are $z_1 = 3+i$ $z_2 = 3-i$



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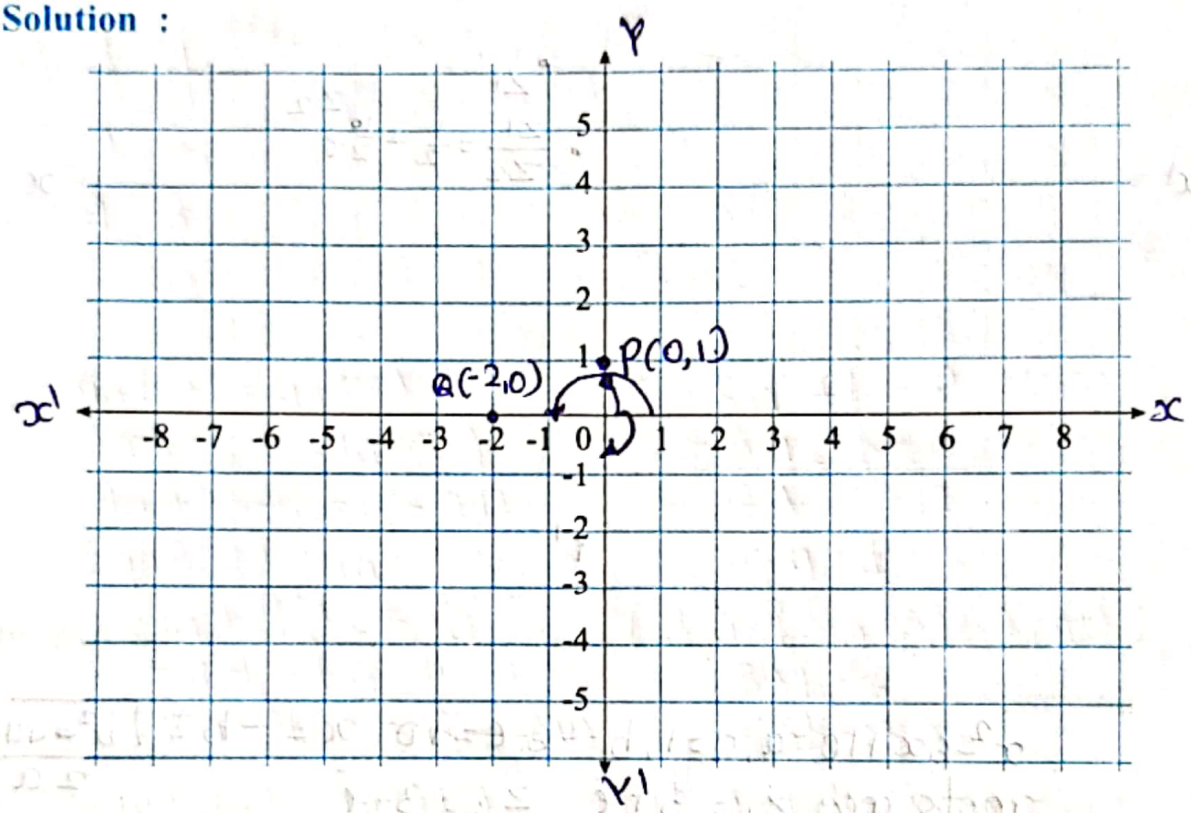
5) By means of Argand diagram, for $z_1 = 0 + i$ and $z_2 = -2+0i$, verify the following :

(a) $|z_1 . z_2| = |z_1| . |z_2|$ (b) $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$

(c) $\arg(z_1.z_2) = \arg z_1 + \arg z_2$ (d) $\arg\left(\frac{z_1}{z_2}\right) = arg(z_1) - arg(z_2)$

Solution :



a) $|z_1| = |0+i| = \sqrt{1} = 1$

$|z_2| = |-2+0i| = \sqrt{4} = 2$

$|z_1 . z_2| = |0-2i| = \sqrt{4} = 2$

$|z_1||z_2| = (1)(2) = 2$

$\therefore |z_1 z_2| = |z_1||z_2|$

Verification using Argand Diagram

$|z_1| = |0+i| = l(OP) = 1$

$|z_2| = |-2+0i| = l(OQ) = 2$

$|z_1 z_2| = |0-2i| = l(OR) = 2$

$|z_1 z_2| = |z_1||z_2| = (1)(2) = 2$

b) $\frac{z_1}{z_2} = \frac{0+i}{-2+0i} = 0 - (\frac{1}{2})i$

Point A represent z_1

Point Q represent z_2

Point R represent $\frac{z_1}{z_2}$

$|z_1| = |0+i| = l(OA) = 1$

$|z_2| = |-2+0i| = l(OB) = 2$

$\left|\frac{z_1}{z_2}\right| = |0 - \frac{1}{2}i| = l(OC) = \frac{1}{2}$

$\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|} = \frac{1}{2}$

(c) $\arg(z_1)$ = angle made by OP with +ve x axis $= \pi/2$

$\arg(z_2)$ = angle made by OQ with +ve x axis $= \pi$

$\arg(z_1) + \arg(z_2) = \frac{\pi}{2} + \pi = 3\pi/2$

$= \arg(z_1 z_2)$

(d) $\arg\left(\frac{z_1}{z_2}\right)$ angle made by OR with +ve x axis $= -\pi/2$

$\arg(z_1) - \arg(z_2) = \frac{\pi}{2} - \pi = -\pi/2$

$= \arg\left(\frac{z_1}{z_2}\right)$

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6) Find square roots of $-7 + 24i$.

Solution : Let $\sqrt{-7+24i} = x+yi$, where $x, y \in R$

an squarring both sides we get

$\rightarrow +24i = (x+yi)^2 = x^2+2xyi+y^2i^2$

$\rightarrow +24i = x^2-y^2+2xyi$

Equating the real and imaginary parts Separately

$x^2-y^2=-7$, $2xy=24$ $\therefore y=\frac{12}{x}$

$\therefore x^2-\left(\frac{12}{x}\right)^2=-7$

$\therefore x^2-\frac{144}{x^2}=-7$

$\therefore x^4-144=-7x^2$

$\therefore x^4+7x^2-144=0$

$\therefore x^4+16x^2-9x^2-144=0$

$\therefore x^2(x^2+16)-9(x^2+16)=0$

$\therefore (x^2+16)(x^2-9)=0$

$\therefore x^2=-16$, $x^2=9$

Now x is real number

$\therefore x^2 \neq -16$

$\therefore x^2=9$

$\therefore x=\pm 3$

when $x=3$, $y=\frac{12}{3}=4$

when $x=-3$, $y=\frac{12}{-3}=-4$

$\therefore$ the square root of $-7+24i$ are

$3+4i$ and $-3-4i$

i.e. $\pm(3+4i)$

Sign of Teacher :

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